MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF AERONAUTICAL AND AUTOMOBILE ENGINEERING			
END SEMESTER EXAMINATION		**Dec 2024**	
Level	**B.Tech**	**Curriculum**	**2022**
Program Name	**Automobile Engineering**	**Course Type**	**C**
Course Name	**Material science and Metallurgy**	**Course Code**	**AAE 2127**
Semester	**III**	**Max Marks**	**50**
Date	**02-12-2024**	**Duration**	**3 Hrs**

Instructions:
- **Answer ALL the questions.**
- **Data not provided may be suitably assumed.**

Q. No.	Question	Marks	NBA		IET*		BL
			CO	PO	CLO	AHEP4 LO	
1A	Compare flame hardening and induction hardening methods, mentioning at least two advantages of each.	**3**	4	1,2,5	4	**1,5,13**	**4**
1B	Differentiate different types of steels in terms of carbon content and their typical applications.	**2**	5	1,2,5	5	**1,4,13**	**4**
1C	Melting temperatures of Metal A and Metal B are 1150 ^{0}C and 750 ^{0}C respectively. The metals A and B are mutually soluble in the liquid state and partly soluble in the solid state. A liquid phase alloy containing 40 % A and 60% B completely transforms into a mixture of two solid solutions at 450 ^{0}C. Maximum solubility of A in B and B in A are 20% and 10% respectively at 450^{0}C, 10% and 5% at room temperature. Assuming the solubility lines to be linear, draw a phase diagram to scale and label the regions. For 15% A alloy determine the following: a) Weight percentage of eutectic formed	**5**	3	1,2,4	3	**2,3,13**	**5**

	b) Temperature where equal proportions of liquid and solid phases exist						
2A	List the point's defects, and explain any two with neat sketches.	**3**	1	1,2	1	**1,12**	**2**
2B	Show the miller indices for the following (1 1 0), (2 0 0), [0 2 1].	**3**	1	1	1	**1,3**	**3**
2C	Explain the Jominy end-quench test and how it is used to determine the hardenability of steel.	**4**	4	1,4,5	4	**1,12,13**	**2**
3A	Explain the favourable conditions for the formation of interstitial solid solution.	**4**	2	1,2,5	2	**1,13**	**2**
3B	Find the atomic packing factor for the FCC unit cell	**3**	1	1	1	**1,2,3**	**3**
3C	With a neat sketch explain grain boundaries and tilt boundaries.	**4**	1	1,2	1	**1,12**	**2**
4A	Differentiate crystalline and amorphous solids.	**3**	1	1,2	1	**1,13**	**4**
4B	With a neat sketch explain the steps involved and process of the TTT (Time-Temperature-Transformation) diagram in understanding phase transformations of steel.	**5**	3	1,2,4	3	**1,5,13**	**2**
4C	List different types of composite materials.	**2**	2	1,2	2	**1,4,13**	**2**
5A	Define heat treatment and list three primary objectives of this process.	**3**	4	1,5	4	**1,5,13**	**4**
5B	List and explain the Hume-Rothery rules.	**2**	1	1,2	1	**1,2**	**2**
5C	Draw the iron-iron carbide (Fe-Fe_3C) phase diagram and label key regions, including phases, invariant reactions, and critical temperatures.	**5**	3	1,4	3	**1,2,3**	**5**

Retain only one. Delete the others

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	***18***
2	***5***
3	***15***
4	***10***
5	***2***

BL	Total Marks
2	26
3	6
4	8
5	10